

Activity: Create a test plan for each of your block's interfaces.

Block 1: Voltage Regulator

Adapter and Voltage Regulator Configuration and Verification:

First, verify the stability of the input interface via the barrel connector. Connect a DC power supply to the board's input terminal, and attach an electronic load to the circuit's main power line to simulate the load of the entire system. Place the probes of a digital multimeter directly onto the input pin of the barrel connector to monitor the voltage supplied by the adapter in real time. The verification process is conducted across four corner cases:

First, test the system for 30 seconds under an input voltage of 12.6V and a load corresponding to the system's nominal current of 1.08A to ensure the voltage remains within the 11.4V to 12.6V range. (Note: The nominal current here represents the sum of the 1A drawn by the two motors and the 80mA drawn by the microcontroller.)

Second, at the same input voltage, increase the load to the system's peak current of 2.25A and verify that the system can sustain this load for 3 seconds.

Third, lower the input voltage to 11.4V and take measurements for 30 seconds under a 1.08A load.

Fourth—and finally—apply the peak load of 2.25A for 3 seconds at an input voltage of 11.4V.

To pass this verification process, the barrel connector and input wiring must maintain a stable voltage range throughout these steps without exhibiting any voltage drop.

Motor Driver Output Configuration and Verification:

Once the input-side verification is complete, inspect the power path leading to the motor driver. Connect the electronic load across the 470 μ F bulk capacitor located at the motor output terminals, and place a digital multimeter in parallel with it. Since the current circuit design does not incorporate an input protection diode, the measured output voltage should be nearly identical to the input voltage. First, with a 12.6V input and the motor operating at its nominal load of 1.0A, conduct a 30-second test to verify that the voltage remains within the range of 11.4V to 12.6V.

Second, with a 12.6V input, apply the motor's peak current—2.0A—for 3 seconds to check for voltage stability.

Third and fourth, lower the input voltage to 11.4V and repeat the same nominal and peak load tests.

In all cases, the voltage must not drop below 11.4V.

Microcontroller Output Configuration and Verification:

Finally, verify the power path for the control section. Connect an electronic load across the 330 μ F capacitor at the output terminal of the LM2576-5.0WT buck converter, and monitor the voltage using a digital multimeter.

First, with a 12.6V input and the control section operating at its nominal load of 80mA, take measurements for 30 seconds to verify that the voltage is precisely maintained within the range of 4.8V to 5.2V.

Second, with a 12.6V input, apply the control section's peak current—250mA—for 3 seconds.

Third and fourth, lower the input voltage to 11.4V and perform the tests under the same nominal and peak load conditions.

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Block 1: Voltage Regulator

1. Required Equipment

Digital Multimeter (DMM), 12V 8A Wall Adapter, 2 Electronic Load

Setup:

1. Disconnect the provided PCB (Power Unit) from the MCU, motor driver, and motor.
2. Plug the 12V 8A wall adapter into a wall outlet and connect the DC barrel plug to the barrel connector on the PCB.
3. Set the DMM to DC voltage measurement mode.

Test:

1. Connect the DMM's red probe to the anode of the Schottky diode and the black probe to ground.
2. Verify on the DMM that the voltage falls within the acceptable range of V_{max} : 12.6V and V_{min} : 11.4V.
3. If the voltage falls within this range, the test passes; otherwise, it fails.

3. Setup and Input Interface Verification

Setup: Keep the adapter connected and prepare the electronic load equipment.

Connect two electronic loads. For the first load, connect the red probe between the Schottky diode and the electrolytic capacitor. Connect the second electronic load to the Vout pin of the voltage regulator. For both loads, connect the black probe to GND.

Case 1) MCU_Ipeak, MD_Inom

Set the first electronic load to the MCU's Ipeak value of 500 mA, and set the second electronic load to the MD's nominal current (Inom) of 4 A.

Case 2) MCU_Ipeak, MD_Ipeak